



UNIVERSITY OF SAN-JOSE RECOLETOS

Engineering Mathematical Society

A.Y. 2026 - 2027



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“LIMITS OF FUNCTIONS”

Practice Problems

1. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$
2. $\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{x}$
3. $\lim_{x \rightarrow 1} \frac{x^3 + 2x^2 - x - 2}{x - 1}$
4. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$
5. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$
6. $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$
7. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$
8. $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$
9. $\lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1}$
10. $\lim_{x \rightarrow 2} \frac{x^3 - 4x}{x - 2}$



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“LIMITS OF FUNCTIONS”
Step-By-Step Solutions

1. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - x - 2}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.
- **Step 1:** Factor the numerator and denominator.

$$\Rightarrow \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)(x+1)}$$

- **Step 2:** Cancel the common factor and substitute $x = 2$.

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x+2}{x+1}$$

- **Step 3:** Simplify.

$$\Rightarrow \lim_{x \rightarrow 2} \frac{2+2}{2+1}$$

Answer: $\frac{4}{3}$

2. $\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{x}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.
- **Step 1:** Expand the numerator and simplify.

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^2 + 4x + 4 - 4}{x}$$



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$$\text{➤ } \lim_{x \rightarrow 0} \frac{x^2 + 4x}{x}$$

- **Step 2:** Factor out x to cancel.

$$\text{➤ } \lim_{x \rightarrow 0} \frac{x(x+4)}{x}$$

- **Step 3:** Substitute $x = 0$ and simplify.

$$\text{➤ } \lim_{x \rightarrow 0} x + 4$$

$$\text{➤ } \lim_{x \rightarrow 0} 0 + 4$$

Answer: 4

$$3. \lim_{x \rightarrow 1} \frac{x^3 + 2x^2 - x - 2}{x - 1}$$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.
- **Step 1:** Factor the numerator.

$$\text{➤ } \lim_{x \rightarrow 1} \frac{(x-1)(x^2 + 3x + 2)}{x - 1}$$

- **Step 2:** Cancel the common factor and substitute $x = 1$.

$$\text{➤ } \lim_{x \rightarrow 1} x^2 + 3x + 2$$

- **Step 3:** Simplify.

$$\text{➤ } \lim_{x \rightarrow 1} 1^2 + 3(1) + 2$$

Answer: 6



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4. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.

- **Step 1:** Factor using the difference of cubes. $\Rightarrow a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x-2}$$

- **Step 2:** Cancel the common factor and substitute $x = 2$.

$$\Rightarrow \lim_{x \rightarrow 2} x^2 + 2x + 4$$

- **Step 3:** Simplify.

$$\Rightarrow \lim_{x \rightarrow 2} 2^2 + 2(2) + 4$$

Answer: 12

5. $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.

- **Step 1:** Factor using the difference of squares. $\Rightarrow a^2 - b^2 = (a - b)(a + b)$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2}$$

- **Step 2:** Cancel the common factor and substitute $x = 2$.

$$\Rightarrow \lim_{x \rightarrow 2} x + 2$$



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- **Step 3:** Simplify.

$$\triangleright \lim_{x \rightarrow 2} 2 + 2$$

Answer: 4

6. $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.
- **Step 1:** Factor using the difference of cubes. $\Rightarrow a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$\triangleright \lim_{x \rightarrow 3} \frac{(x-3)(x^2+3x+9)}{(x-3)}$$

- **Step 2:** Cancel the common factor and substitute $x = 3$.

$$\triangleright \lim_{x \rightarrow 3} x^2 + 3x + 9$$

- **Step 3:** Simplify.

$$\triangleright \lim_{x \rightarrow 3} 3^2 + 3(3) + 9$$

Answer: 27

7. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.
- **Step 1:** Factor using the difference of squares. $\Rightarrow a^2 - b^2 = (a - b)(a + b)$



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$$\triangleright \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{(x-4)}$$

- **Step 2:** Cancel the common factor and substitute 4.

$$\triangleright \lim_{x \rightarrow 4} x + 4$$

- **Step 3:** Simplify.

$$\triangleright \lim_{x \rightarrow 4} 4 + 4$$

Answer: 8

8. $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.

- **Step 1:** Factor using the difference of squares. $\Rightarrow a^2 - b^2 = (a - b)(a + b)$

$$\triangleright \lim_{x \rightarrow -2} \frac{(x-2)(x+2)}{x+2}$$

- **Step 2:** Cancel the common factor and substitute $x = -2$.

$$\triangleright \lim_{x \rightarrow -2} x - 2$$

- **Step 3:** Simplify.

$$\triangleright \lim_{x \rightarrow -2} (-2) - 2$$

Answer: - 4



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9. $\lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.

- **Step 1:** Factor the numerator.

$$\Rightarrow \lim_{x \rightarrow 1} \frac{(x+4)(x-1)}{x-1}$$

- **Step 2:** Cancel the common factor and substitute $x = 1$.

$$\Rightarrow \lim_{x \rightarrow 1} x + 4$$

- **Step 3:** Simplify.

$$\Rightarrow \lim_{x \rightarrow 1} 1 + 4$$

Answer: 5

10. $\lim_{x \rightarrow 2} \frac{x^3 - 4x}{x - 2}$

- **Rule:** First, substitute the approaching value. If the result is $\frac{0}{0}$, factor the numerator and denominator, cancel the common factor, then substitute again.

- **Step 1:** Factor the numerator by taking out the common factor x .

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x(x^2 - 4)}{x - 2}$$

- **Step 2:** Then, use the difference of squares. $\Rightarrow a^2 - b^2 = (a - b)(a + b)$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x(x-2)(x+2)}{x-2}$$

- **Step 3:** Cancel the common factor and substitute $x = 2$.



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➤ $\lim_{x \rightarrow 2} x(x + 2)$

- **Step 3:** Simplify.

➤ $\lim_{x \rightarrow 2} 2(2 + 2)$

Answer: 8